

#HW 4

● Graded

Student

Jacky Shen

Total Points

88 / 100 pts

Question 1

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 2 pts quality not found out.

- 2 pts The final state temperature is not correct.

- 2 pts Quality found out is not correct.

Question 2

(no title)

10 / 10 pts

✓ - 0 pts Correct

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts The work calculation is wrong.

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

Question 5

(no title)

8 / 10 pts

- 0 pts Correct

- 2 pts correct answer 71 degree C approx.

✓ - 2 pts solution not shown.

Question 6

(no title)

7 / 10 pts

– 0 pts Correct

✓ – 3 pts incomplete

Question 7

(no title)

10 / 10 pts

✓ – 0 pts Correct

– 5 pts incomplete

Question 8

(no title)

6 / 10 pts

– 0 pts Correct

– 2 pts Correct answer:0.95 cubic m

✓ – 4 pts incomplete

Question 9

(no title)

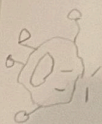
7 / 10 pts

– 0 pts Correct

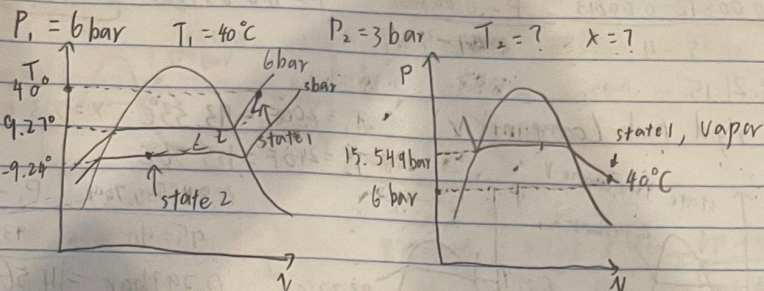
– 2 pts Correct answer: 10 bar approx.

✓ – 3 pts incomplete

Questions assigned to the following page: [1](#) and [2](#)



1. 3.21.14



specific volume at 6 bar 40°C is $0.24118 \text{ m}^3/\text{kg}$

new $v = 0.24118$, $P = 3 \text{ bar}$, T_2

since v is bigger than 1.5361×10^{-3} , smaller than 0.4061 , assume it's between

$$0.24118 = (1-x)(1.5361 \times 10^{-3}) + x(0.4061)$$

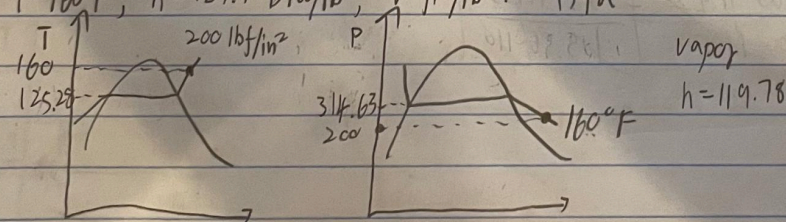
$$0.24118 = 1.5361 \times 10^{-3} - 1.5361 \times 10^{-3}x + 0.4061x$$

$$0.2396439 = 0.4045639x$$

$$x = 0.59, T_2 = -9.24^\circ\text{C}$$

2. 3.21.29

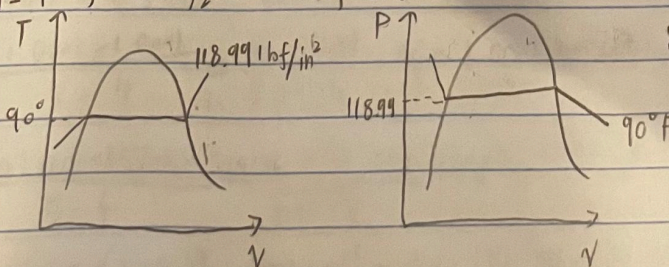
(a) $T = 160^\circ\text{F}$, $h = 127.7 \text{ Btu/lb}$, $v \text{ ft}^3/\text{lb}$?



since higher than evaporated vapor, it's gas

$$P = 200 \text{ lbf/in}^2, v = 0.2636 \text{ ft}^3/\text{lb}$$

(b) $T = 90^\circ\text{F}$, $u = 72.21 \text{ Btu/lb}$, h ?



since u is between u_f and u_g ,

it's mixture

$$72.21 = (1-x)(40.42) + x(105)$$

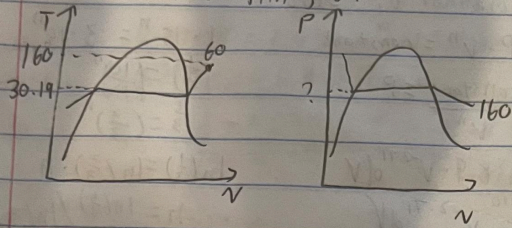
$$x = 0.49$$

$$(1-0.49)(40.72) + (0.49)(113.75)$$

$$= 76.5047 \text{ Btu/lb}$$

Question assigned to the following page: [2](#)

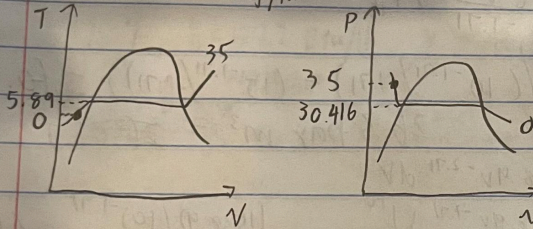
c) $T = 160^\circ\text{F}$ $P = 60 \text{ lbf/in}^2$, u ?



vapor

$$u = 624.44 \text{ Btu/lb}$$

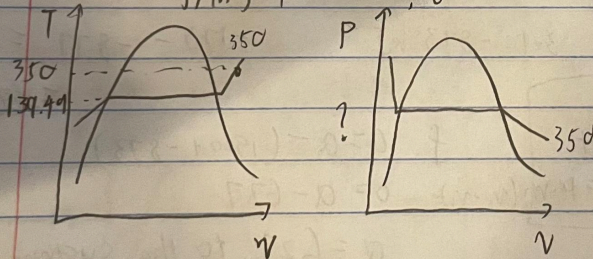
d) $T = 0^\circ\text{F}$ $P = 35 \text{ lbf/in}^2$, u ?



liquid

$$u = 42.32 \text{ Btu/lb}$$

e) $P = 350 \text{ lbf/in}^2$, $T = 350^\circ\text{F}$, u ?



vapor

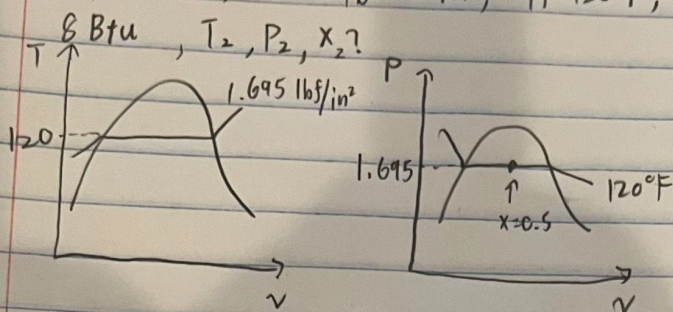
$$\frac{144.95 - 141.32}{360 - 340} = \frac{x - 141.32}{350 - 340}$$

$$u = 143.135 \text{ Btu/lb}$$

Questions assigned to the following page: [3](#) and [4](#)

4 3.21.46

rigid tank, 0.02 lb water, $T_1 = 120^\circ\text{F}$, $x = 0.5$, $P_1 = 1.695$



$$8 / 0.02 = 400 \text{ Btu/lb}$$

$$U = U_1 + 400 \quad U_1 = (1-x)(87.99) + (0.5)(1049.9)$$

$$U = 968.945 \quad U_1 = 568.945$$

$$968.945 = (1-x)(87.99) + x(1049.9)$$

$$T_2 = 120^\circ\text{F}, P_2 = 1.695 \text{ lbf/in}^2, x = 0.92$$

3. 3.21.36

constant P, from saturated vapor at 4 bar to 30°C

$$P_1 = 4 \text{ bar} \quad T_2 = 30^\circ\text{C}$$

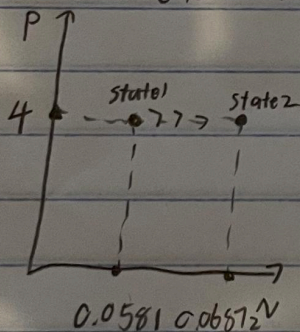
$$T_1 = -6.56^\circ\text{C} \quad P_2 = 4 \text{ bar}$$

p-v diagram, work, Q? $c. 0.06812$

$$W = \int P dv = P(v_2 - v_1) = P \cdot m(v_2 - v_1)$$

$$W = 0.04248 \text{ bar m}^3 = 4.248 \text{ kJ/kg}$$

$$\Delta U = (245.73 - 224.24) = 21.49 = Q - 4.248 \quad Q = 25.738$$



5. 18.7.6

